

CSS ::before and ::after

Pseudo-Elements Mastery Guide

Master the most powerful CSS pseudo-elements for design

Easy English • 50+ Examples • Ready to Use

Contents

Chapter 1

Fundamentals of `::before` and `::after`

1.1 What are `::before` and `::after`?

The `::before` and `::after` pseudo-elements are virtual elements that CSS creates. They allow you to add content before or after an element without adding extra HTML tags. These are among the most powerful CSS features for creating decorative effects and layouts.

Think of it as: "Inject virtual content anywhere on your page using pure CSS."

Key Concept

`::before` and `::after` create virtual child elements that can be styled independently. They are extremely useful for decorations, icons, overlays, and more—all without touching your HTML.

1.2 The Content Property

The `content` property is required for `::before` and `::after` to display:

```
element::before {  
    content: "text";  
}  
  
element::after {  
    content: "";  
}
```

1.3 Order of Elements

When you add `::before` and `::after` to an element, they arrange like this:

`::before` content → **Element Content** → `::after` content

If an element already has content (like text in a `<p>` tag), `::before` and `::after` will wrap around it.

Chapter 2

Mastering the Content Property

2.1 1. Text Content

2.1.1 Basic Text

<

HTML:

b>This is important</p>

Result: "Note: This is important"

2.1.2 Multiple Lines

CSS Code

```
p::before {  
  content: "Line 1\\ALine 2";  
  white-space: pre;  
}
```

2.2 2. Emoji Content

CSS Code

```
li::before {
    content: " ";
    color: green;
    font-weight: bold;
    margin-right: 8px;
}

button::before {
    content: " ";
    margin-right: 6px;
}

a[target="_blank"]::after {
    content: " ";
    font-size: 0.8em;
}
```

2.3 3. Unicode Characters

Unicode characters provide symbols without emojis:

```
]
```

```
/* Checkmark */
li::before {
    content: "\2714";
}

/* Bullet */
li::before {
    content: "\2022";
}

/* Copyright */
p::after {
    content: " \A9 2024";
}

/* Arrow right */
a::after {
    content: " \2192";
}

/* Star */
span::before {
    content: "\2605";
    color: gold;
}
```

```
]

/* Show URL after links */
a::after {
    content: " (" attr(href) ")";
    font-size: 0.8em;
    color: gray;
}

/* Show alt text */
img::after {
    content: attr(alt);
    display: block;
    text-align: center;
}

/* Show title attribute */
abbr::after {
    content: " (" attr(title) ")";
    font-size: 0.9em;
}

/* Data attributes */
div::after {
    content: attr(data-tooltip);
}
```

<

HTML:

```
b href="/page.html">Link</a> <abbr title="Application Programming Interface">API</abbr>
```

Result: - Link displays as: "Link (/page.html)" - API displays as: "API (Application Programming Interface)"

2.4 5. Empty Content

Sometimes you want the pseudo-element for styling only:

```
]

/* Create a decorative line */
h2::after {
  content: "";
  display: block;
  height: 2px;
  background-color: blue;
  margin-top: 10px;
}

/* Create a gradient background overlay */
.image::after {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background: linear-gradient(
    to bottom,
    rgba(0, 0, 0, 0),
    rgba(0, 0, 0, 0.5)
  );
}
```

```
]

.container::after {
  content: "";
  display: table;
  clear: both;
}
```

```
]

button {
  padding: 10px 20px;
}

button.search::before {
  content: " ";
  margin-right: 6px;
}

button.download::before {
  content: " ";
  margin-right: 6px;
}

button.close::after {
  content: " ";
```



```
margin-left: 6px;
}
```

<

HTML:

```
button class="search">Search</button> <button class="download">Download</button>
<button class="close">Close</button>
```

2.5 Pattern 3: Link Indicators

Show different icons based on link type:

]

```
/* External links */
a[href^="http"]::after {
    content: " ";
    font-size: 0.8em;
    margin-left: 3px;
}

/* Email links */
a[href^="mailto:"]::before {
    content: " ";
}

/* Phone links */
a[href^="tel:"]::before {
    content: " ";
}

/* PDF downloads */
a[href$=".pdf"]::after {
    content: " ";
}

/* Video links */
a[href$=".mp4"]::before {
    content: " ";
}
```

]

```
.image-container {
    position: relative;
    overflow: hidden;
}

.image-container::after {
    content: "";
    position: absolute;
    top: 0;
    left: 0;
    right: 0;
    bottom: 0;
}
```

```
background: linear-gradient(
    135deg,
    rgba(255, 0, 0, 0.3) 0%,
    rgba(0, 0, 255, 0.3) 100%
);
pointer-events: none;
}
```

]

```
/* Line under headings */
h2::after {
    content: "";
    display: block;
    height: 3px;
    background: linear-gradient(
        to right,
        blue,
        transparent
    );
    margin-top: 10px;
}
```

```
/* Vertical divider */
.divider::before,
.divider::after {
    content: "";
    display: inline-block;
    width: 1px;
    height: 20px;
    background-color: #ccc;
    margin: 0 15px;
    vertical-align: middle;
}
```

```
/* Dashed border */
.box::before {
    content: "";
    position: absolute;
    top: 0;
    left: 0;
    right: 0;
    bottom: 0;
    border: 2px dashed blue;
    pointer-events: none;
}
```

]

```
/* Underline animation */
a {
    position: relative;
    text-decoration: none;
}
```

```
a::after {
    content: "";
    position: absolute;
    width: 100%;
    height: 2px;
    bottom: -5px;
    left: 0;
    background-color: blue;
    transform: scaleX(0);
    transform-origin: bottom right;
    transition: transform 0.3s ease;
}
```

```
a:hover::after {
    transform: scaleX(1);
    transform-origin: bottom left;
}
```

/* Background slide effect */

```
button {
    position: relative;
    overflow: hidden;
}
```

```
button::before {
    content: "";
    position: absolute;
    top: 0;
    left: -100%;
    width: 100%;
    height: 100%;
    background-color: darkblue;
    transition: left 0.3s ease;
    z-index: -1;
}
```

```
button:hover::before {
    left: 0;
}
```

]

```
.breadcrumb li {
    display: inline-block;
}
```

```
.breadcrumb li::before {
    content: " / ";
    margin: 0 8px;
    color: gray;
}
```

```
.breadcrumb li:first-child::before {
    content: "";
}
```

```
/* Alternative with arrows */
.breadcrumb-arrow li::after {
    content: " \2192 ";
    margin: 0 8px;
}

.breadcrumb-arrow li:last-child::after {
    content: "";
}
```

<

HTML:

```
bl class="breadcrumb"> <li>Home</li> <li>Products</li> <li>Electronics</li> <li>Phones</li>
</ol>
```

Chapter 3

Practical Projects

3.1 Project 1: Custom List Styling

HTML

```
<ul class="custom-list">
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ul>
```

```
.custom-list {
  list-style: none;
  padding: 0;
}

.custom-list li {
  padding: 10px 0 10px 30px;
  position: relative;
}

.custom-list li::before {
  content: "";
  position: absolute;
  left: 0;
  color: green;
  font-weight: bold;
  font-size: 1.2em;
}

.custom-list li:hover::before {
  transform: scale(1.3);
  transition: transform 0.2s ease;
}
```

```
<span class="tooltip" data-tooltip="Click to expand">
  Hover me
</span>
```

```
]

.tooltip {
  position: relative;
  cursor: help;
  border-bottom: 1px dotted blue;
}

.tooltip::after {
  content: attr(data-tooltip);
  position: absolute;
  bottom: 125%;
  left: 50%;
  transform: translateX(-50%);
  background-color: #333;
  color: white;
  padding: 8px 12px;
  border-radius: 4px;
  white-space: nowrap;
  z-index: 1;
  opacity: 0;
  pointer-events: none;
  transition: opacity 0.3s ease;
}

.tooltip::before {
  content: "";
  position: absolute;
  bottom: 115%;
  left: 50%;
  transform: translateX(-50%);
  border: 5px solid transparent;
  border-top-color: #333;
  opacity: 0;
  transition: opacity 0.3s ease;
}

.tooltip:hover::after,
.tooltip:hover::before {
  opacity: 1;
}
```

```
]

<div class="card" data-badge="New">
  
  <h3>Product Name</h3>
  <p>Product description</p>
</div>
```

```
]
```

```
.card {
  position: relative;
  width: 250px;
  background: white;
  border-radius: 8px;
  overflow: hidden;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
}

.card::before {
  content: attr(data-badge);
  position: absolute;
  top: 10px;
  right: 10px;
  background-color: red;
  color: white;
  padding: 5px 10px;
  border-radius: 20px;
  font-size: 0.8em;
  z-index: 10;
}

.card::after {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background: linear-gradient(
    135deg,
    rgba(255, 255, 255, 0.2) 0%,
    transparent 50%
  );
  pointer-events: none;
}
```

]

```
<button class="btn-loading">Send</button>
```

]

```
button {
  padding: 10px 20px;
  background-color: blue;
  color: white;
  border: none;
  cursor: pointer;
}

button:active {
  opacity: 0.8;
}
```

```
button::after {
  content: "";
  margin-left: 8px;
}

button.loading::after {
  content: "";
  display: inline-block;
  animation: spin 1s linear infinite;
}

@keyframes spin {
  from { transform: rotate(0deg); }
  to { transform: rotate(360deg); }
}
```

```
<span class="price" data-original="100">
  \ $75
</span>
```

```
.price {
  position: relative;
  font-weight: bold;
}

.price::before {
  content: "Original: \ $" attr(data-original);
  position: absolute;
  top: -25px;
  left: 0;
  font-size: 0.8em;
  color: gray;
  text-decoration: line-through;
}

.price::after {
  content: " 25% OFF";
  background-color: red;
  color: white;
  padding: 2px 6px;
  border-radius: 3px;
  font-size: 0.7em;
  margin-left: 5px;
}
```

```
a {
  position: relative;
  text-decoration: none;
}
```



```
a::after {
  content: "";
  position: absolute;
  width: 0%;
  height: 2px;
  bottom: -2px;
  left: 0;
  background-color: blue;
  transition: width 0.3s ease;
}
```

```
a:hover::after {
  width: 100%;
}
```

]

```
.box {
  position: relative;
  width: 200px;
  height: 150px;
}

.box::before {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  border: 2px solid blue;
  border-radius: 4px;
  transform: scale(0.95);
  opacity: 0;
  transition: all 0.4s ease;
}
```

```
.box:hover::before {
  transform: scale(1);
  opacity: 1;
}
```

]

```
button::after {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: linear-gradient(
    90deg,
    transparent,
    rgba(255, 255, 255, 0.3),

```

```
        transparent
    );
    transform: translateX(-100%);
    transition: transform 0.6s ease;
}

button:hover::after {
    transform: translateX(100%);
}
```

```
    ]

ol {
    counter-reset: item;
    list-style: none;
    padding-left: 0;
}

ol li {
    counter-increment: item;
    margin-bottom: 10px;
    position: relative;
    padding-left: 30px;
}

ol li::before {
    content: counter(item);
    position: absolute;
    left: 0;
    top: 0;
    width: 24px;
    height: 24px;
    background-color: blue;
    color: white;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 0.9em;
}
```

```
    ]

blockquote {
    position: relative;
    padding: 20px;
    border-left: 4px solid blue;
    margin: 20px 0;
    font-style: italic;
}

blockquote::before {
    content: "\201C";
    font-size: 4em;
    color: rgba(0, 0, 0, 0.1);
}
```

```
    position: absolute;
    top: -20px;
    left: 10px;
}

blockquote::after {
    content: "\201D";
    font-size: 4em;
    color: rgba(0, 0, 0, 0.1);
    float: right;
    margin-top: 20px;
}
```

```
    ]

label::after {
    content: " *";
    color: red;
}

input[required]::after {
    content: "";
    display: inline-block;
    width: 8px;
    height: 8px;
    background-color: red;
    border-radius: 50%;
    margin-left: 4px;
}
```

/

Chapter 4

Tips and Best Practices

4.1 Tip 1: Always Include content Property

b WRONG - Nothing displays */ p::before color: red; font-size: 1.5em;
/* RIGHT - Must have content */ p::before content: "Important: "; color: red; font-size: 1.5em;

4.2 Tip 2: Use position: relative

When positioning ::before and ::after absolutely, ensure the parent has `position: relative`:

```
.container {  
    position: relative;  
}  
  
.container::after {  
    position: absolute;  
    top: 0;  
    left: 0;  
}
```

4.3 Tip 3: Use z-index Correctly

Control layering of pseudo-elements:

```
element::before {  
    z-index: 1;  
}  
  
element {  
    z-index: 2;  
}  
  
element::after {  
    z-index: 3;  
}
```

4.4 Tip 4: Use pointer-events: none

Prevent pseudo-elements from blocking interactions:

```
element::after {  
    pointer-events: none;  
}
```

4.5 Tip 5: Accessibility Considerations

Don't put essential content in pseudo-elements:

```
/* Good - Decorative */  
button::after {  
    content: " →";  
}  
  
/* Bad - Essential information */  
/* Don't hide important info in pseudo-elements */
```

Chapter 5

Common Mistakes to Avoid

5.1 Mistake 1: Forgetting content Property

```
/* WRONG */
p::before {
    color: red;
}

/* RIGHT */
p::before {
    content: "";
    color: red;
}
```

5.2 Mistake 2: Single Colon Syntax

```
/* OLD (still works but outdated) */
p:before { content: "text"; }

/* NEW (correct) */
p::before { content: "text"; }
```

5.3 Mistake 3: Forgetting position Context

```
/* WRONG - Parent not positioned */
.parent::after {
    position: absolute;
    top: 0;
    left: 0;
}

/* RIGHT - Parent positioned */
.parent {
    position: relative;
}
```

```
.parent::after {  
    position: absolute;  
    top: 0;  
    left: 0;  
}
```

5.4 Mistake 4: Hiding Important Content

```
/* WRONG - Accessibility issue */  
a::after {  
    content: "(opens in new window)";  
    display: none;  
}  
  
/* BETTER - Use title or aria */  
<a href="#" title="Opens in new window">  
    Link  
</a>
```

5.5 Mistake 5: Complex Layouts

```
/* AVOID - Use actual HTML for complex layouts */  
/* ::before and ::after are for decoration/small content */  
  
/* GOOD - Use them for icons/labels */  
button::before {  
    content: " ";  
}
```

Chapter 6

Real-World Examples

6.1 Example 1: Article Timeline

]

```
<ul class="timeline">
  <li>January 2024</li>
  <li>March 2024</li>
  <li>June 2024</li>
</ul>
```

]

```
.timeline {
  list-style: none;
  padding: 20px 0;
}

.timeline li {
  padding-left: 60px;
  position: relative;
  margin-bottom: 30px;
}

.timeline li::before {
  content: "";
  position: absolute;
  left: 0;
  top: 5px;
  width: 20px;
  height: 20px;
  background-color: blue;
  border-radius: 50%;
}

.timeline li::after {
  content: "";
  position: absolute;
  left: 9px;
  top: 30px;
  width: 2px;
```



```
    height: 30px;
    background-color: #ccc;
}

.timeline li:last-child::after {
    display: none;
}
```

```
]
```

```
<div class="rating" data-rating="4.5"></div>
```

```
]
```

```
.rating::before {
    content: "";
    letter-spacing: 3px;
    color: #ddd;
}

.rating[data-rating="5"]::after {
    content: "";
}

.rating[data-rating="4.5"]::after {
    content: "";
}

.rating[data-rating="4"]::after {
    content: "";
}

.rating::after {
    position: absolute;
    left: 0;
    top: 0;
    color: gold;
    letter-spacing: 3px;
}
```